

Nuclear Reactions & Nuclear Power Plant



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Whatever we will discuss

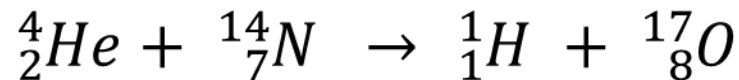
- Nuclear reaction and its types
- Nuclear fission & nuclear fusion with examples
- Nuclear reactor
- Working of a Nuclear power plant

Nuclear Reaction

Nuclear reaction is a process in which two nuclei, or a nucleus and an external subatomic particle, collide to produce one or more new nuclides.

Nuclear reaction, **change in the identity or characteristics of an atomic nucleus, induced by bombarding it with an energetic particle.** The bombarding particle may be an alpha particle, a gamma-ray photon, a neutron, a proton, or a heavy ion.

1st Experimental Nuclear Reaction:



Symbolically, ${}^{14}_7\text{N} (\alpha, p) {}^{17}_8\text{O}$

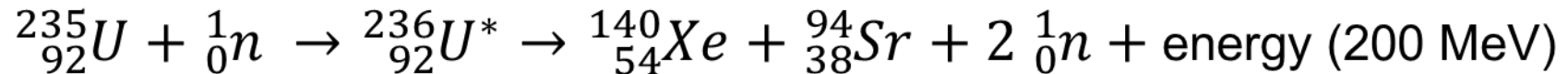
Types of Nuclear reactions-

- 1) Nuclear fission
- 2) Nuclear fusion
- 3) Nuclear decay
- 4) Transmutation

Nuclear Fission & Fusion

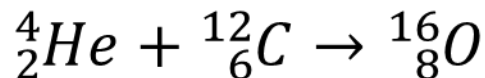
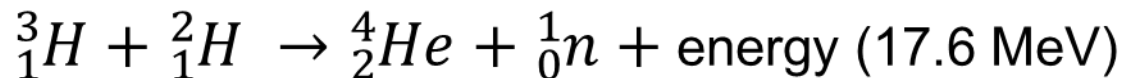
Nuclear Fission:

The process in which a heavy nucleus splits up into two nuclei of nearly comparable masses and emits enormous amount of energy, is called **nuclear fission**.



Nuclear Fusion:

The process by which two or more light nuclei are combined to form a heavy nucleus and enormous energy is released is called **nuclear fusion**.



Chain Reaction:

Chain reaction is such a process which once started the reaction continues without requiring further energy and a tremendous amount of energy is released because of the fission of all nuclei.

Nuclear Reactor

A **nuclear reactor** is a device in which a chain reaction involving nuclear fission can be initiated and controlled.

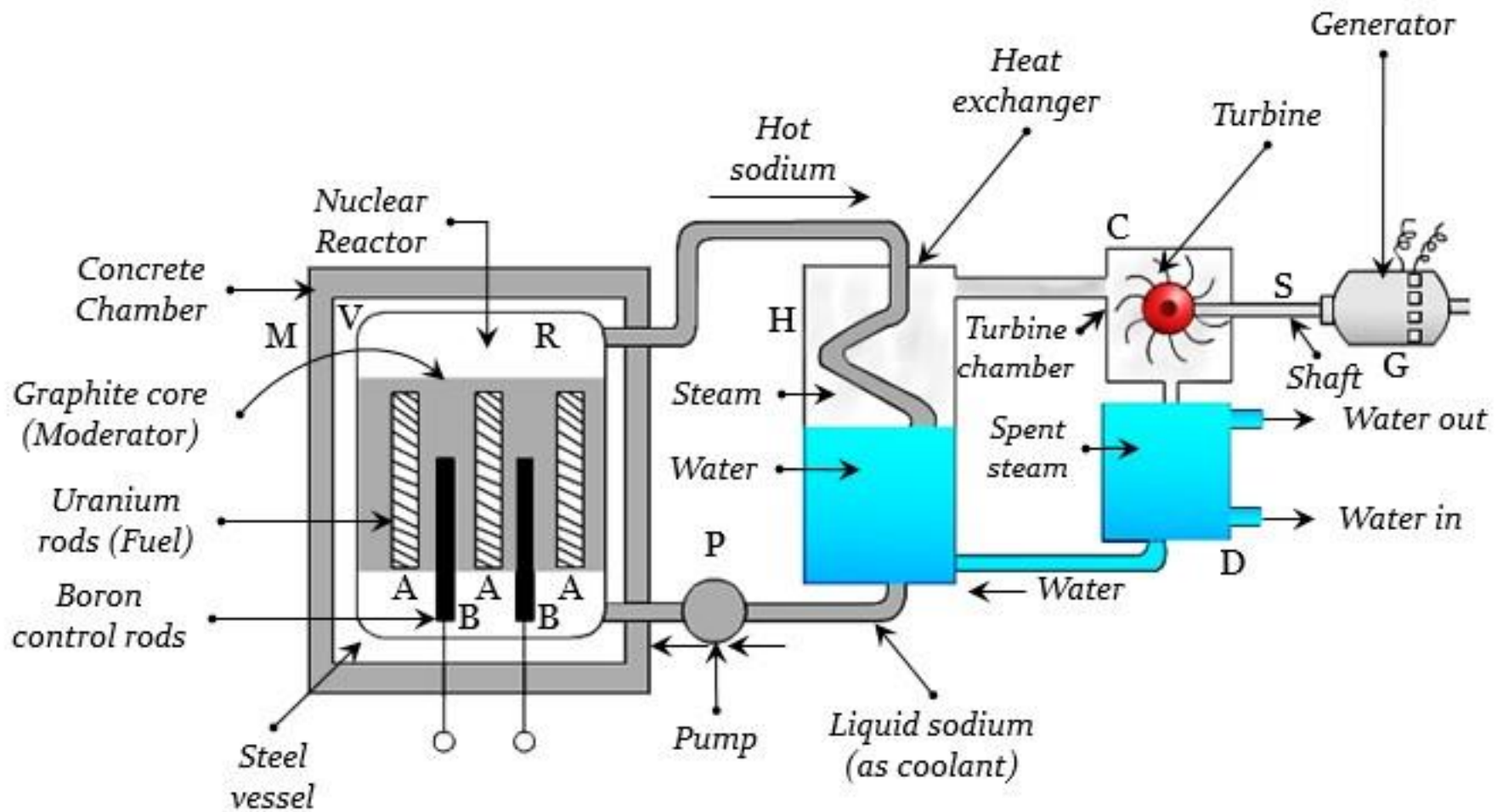
A reactor is a very efficient source of energy: the fission of **1 gm** of a suitable fissionable material per day evolves energy at the rate of **about 1 MW (10^6 watt)**. For producing energy at the same rate combustion of **2.6 tons** of coals per day will be needed.

The energy liberated in a nuclear reactor **becomes heat** in its interior, and this heat is removed by circulating a liquid or gas coolant. The hot coolant is then used to **boil water**, and the resulting **steam is fed to a turbine** that can power an electric generator to **produce electricity**.

The essential elements of a nuclear reactor are-

- 1) The fissionable substance (Uranium),
- 2) Control Rod (Cadmium or Boron),
- 3) Cooling system (Liquid Sodium),
- 4) Safety system (Concrete Chamber)
- 5) Moderator (Graphite or D_2O)
- 6) Neutron reflector

Working of a Nuclear Power Plant





Thank You
Thanks for your attention